Mission 3: PowerSat CubeSat

Mission idea

Students use the CubeSat to simulate how satellites **generate, store, and distribute energy using solar power**. By measuring light input and controlling system outputs, students model how real satellites manage limited power and make decisions about which systems can operate at any given time.

Driving question

How can a CubeSat generate and manage limited energy to keep important systems running in space?

Mission objectives

- Measure how light intensity affects energy generation (simulated solar power)
- Compare system performance under high vs. low energy conditions
- Investigate how increasing power demand affects system behavior
- Make decisions about which subsystems to prioritize when power is limited
- Evaluate how well the CubeSat design manages energy efficiently

Example student activities

- Shine a flashlight vs. dim light on the sensor and record output changes
- Cover the sensor to simulate satellite eclipse (no sunlight)
- Adjust a potentiometer to simulate increased power demand
- Test one vs. multiple “subsystems” (LEDs/buzzer) running at once
- Compare which designs keep systems running the longest under low energy
- Graph light level vs. system output (brightness, activation, etc.)

NGSS alignment

- **MS-PS3-3:** Apply scientific principles to design and test an energy system
- **MS-ETS1-1:** Define the problem of limited energy in a system
- **MS-ETS1-2:** Evaluate design solutions for managing power
- **MS-ETS1-3:** Analyze data to improve energy efficiency
- **MS-ETS1-4:** Develop and refine models of energy systems